

Muhammad Abdullah

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EDUCATION

Information Technology University (ITU), Lahore

Bachelor of Science in Computer Science, CGPA: 3.92/4.00

Aug 2023 – Present

7th Semester

EXPERIENCE

Software Engineering Intern

Jun 2026 – Aug 2026

Powersoft19, Lahore

- Shipped full-stack production features and API integrations, reviewed teammates' PRs, and delivered on sprint cadence.

Computer Vision Intern

Mar 2025 – Jan 2026

AiBee.pk (Hybrid), Lahore

- Fine-tuned YOLOv8 and EfficientDet models for retail analytics, achieving 92% mAP on evaluation sets.
- Optimized TensorRT + Docker inference pipelines, reducing latency by 40% for edge deployment.

Teaching Assistant – Software Engineering

Aug 2025 – Jul 2026

Information Technology University (ITU), Lahore

- Mentored 50+ students in software engineering, Git workflows, and CI/CD practices through labs and code reviews.

Freelance Agentic Systems Engineer

Jan 2025 – Present

Remote

- Replaced one-shot LLM calls with stateful multi-agent graphs (planner → tool-calling workers → verifier), using checkpointed memory so a failed tool retry does not replay the entire run.
- Exposed client APIs as MCP tool servers with A2A-style handoffs, allowlisted writes behind human-in-the-loop gates, and trajectory evals/tracing so actuation is auditable rather than “the model said so.”
- Orchestrated long-running ops workflows across LangGraph, Google ADK, CrewAI, and n8n: durable control flow, typed tool contracts, and policy-bounded recovery instead of chatbot wrappers.

PROJECTS

Agentic SRE | *Multi-Agent Orchestration, Tool-Calling, SSH, Provider APIs*

2026

- Building a multi-agent production runtime for indie SaaS on VPS hosts that investigates incidents with specialist agents (hypotheses, tools, verifier, memory) rather than a single classify-and-done call.
- Designed a governed detect–investigate–plan–contain–remediate–verify loop with allowlisted actuation, policy/human gates, and off-box recovery that does not depend on SSH into a dead guest.

Tomoads AdTech Platform | *Python FastAPI, Go, Next.js, PostgreSQL, ClickHouse, Redis, Cloudflare R2*

- Built and shipped a production ad network stack with advertiser campaign workflows, publisher domain/placement onboarding, and admin controls.
- Implemented secure ad serving and billing flow with signed serve tokens, impression/click tracking, internal debit events, and wallet/ledger payouts.
- Developed reliability and anti-fraud guardrails including domain/path verification integrity, role-based admin permissions, and Redis-backed validation/cache paths.

Make-It-RAG | *Python, Flask, Google Gemini, MySQL, React*

Aug 2025

- Built a schema-aware query agent that plans SQL from natural language, executes it as a tool against the connected database, and retries/self-corrects failed steps in an interactive chat UX.
- Presented the project at the 4th National Artificial Intelligence Seminar (NAIS-2025, National Centre for Physics, Islamabad) in a national competition.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, C++, C#, C, Java, Kotlin, SQL, Assembly

Frameworks/Libraries: React, Next.js, Node.js, FastAPI, Flask, Django, Express, LangChain, LangGraph, CrewAI, AutoGen, Google ADK, LlamaIndex, OpenAI Agents SDK, TensorFlow, PyTorch, Gradio

AI/ML: Agentic AI, Multi-Agent Systems, Tool-Calling, MCP, A2A, Agent Memory, Human-in-the-Loop, Agent Evals, RAG Systems, LLM Integration, Azure OpenAI, Google Gemini, Cohere, Prompt Engineering, CNNs

Databases: PostgreSQL, MongoDB, MySQL, Qdrant, Redis, ClickHouse, Firebase, Supabase

Cloud/DevOps: Docker, Kubernetes, AWS, Azure, GCP, Vercel, GitHub Actions, CI/CD, Terraform

Tools: Git, REST APIs, JWT, RabbitMQ, WebSockets, OpenCV, Microservices, n8n

CERTIFICATIONS

Agentic AI Engineering (*Kaggle, 5-Day Intensive*) Jun 2026

- Focus: Multi-agent systems, tool-calling, agent evaluation, and production-oriented agent workflows.

Microsoft AI & ML Engineering Professional Certificate (*Microsoft*) Jun 2025

- Credential ID: **VOGHSBA4IQ9R** — Focus: Artificial Intelligence, Machine Learning, applied AI engineering workflows.

IBM Certified Database Administrator – DB2 10.5 DBA for LUW Upgrade from DB2 10.1 (*IBM*) Jul 2024

- Credential ID: **8SL22GBC3U7Q** — Focus: Database Administration, DBMS operations, enterprise database reliability.

Advanced AI and Machine Learning Techniques and Capstone (*Microsoft*) May 2025

- Credential ID: **4TWXAIB3BNCW** — Focus: Deep Learning, Artificial Neural Networks, capstone-level AI/ML application.

AWARDS & HONORS

Softec '26 – 2nd Place, App Development Competition 2026

LoopLab '26 – 1st Position, App Development Competition 2026

NAIS '25 Invitee – Make-It-RAG Presentation, National Centre for Physics Jul 2025

PUCon '25 – 1st Place, Application Development Competition 2025

FCCU CODEFEST '25 – 1st Place, Web Hackathon 2025

LoopVerse '25 – 1st Place, Application Development Competition 2025

Softec '25 – 2nd Place, Application Development Challenge 2025